MEMO for: Mr. R. L. Palmer

September 13, 1955

169.12

Subject: STRETCH PROPOSAL TO AEC, LOS ALAMOS

Attached is a draft of the Stretch Proposal which it is planned to present to Los Alamos, on September 20th.

The important differences between this proposal and the Livermore 2 Megacycle Proposal are as follows:

LIVERMORE 2 MC PROPOSAL	LOS ALAMOS 10 MC PROPOSAL
2 MC Transistors	10 MC Transistors
4 200 word 2 Microsecond Fast Core Memories	2 500 word 0.5 Microsecond Fast Core Memories
2 20,000 word 10 Microsecond Large Core Memories	4 10,000 word 2 Microsecond Large Core Memories
Floating Point Add 6 microseconds	Floating Point Add 0.6 microseconds
Multiply 12 microseconds	Multiply 1,2 microseconds
Fast tapes providing 1,200,000 information bits a second (with auto-correction)	Fast tapes providing 2,400,000 information bits a second (auto-correction sacrificed to increase the bit-rate)
Delivery - November 1958 (42 months from Apr. 20, 1955)	Delivery - October 1958 (36 months from Sept. 20, 1955)

The proposal does not state price, and indicates that present standard EDPM input-output devices would be optional. The price and the exact input-output to be included in it would be covered separately in a letter.

Color photographs of several solid state devices built by IBM will be included in two copies of the proposal presented to Los Alamos. They will probably include:

a) The transistorized 604
b) The 704 Core Memory unit
c) A core plane
d) A printed circuit unit

S. W. Dunwell

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SWD:BC
Attach.

cc: Dr. G. M. Amdahl
Mr. M. M. Astrahan
Mr. J. W. Backus
Mr. H. S. Beattie
Dr. W. Bouricius
Mr. J. Griffith ←
Mr. J. A. Haddad
Dr. C. C. Hurd